Thursday, 8 January 2026 6:00 PM

http: → stateless protocol.
- every request coming from a user is treated as new

Problem
- User authentication persistence.
- Personalized data Tracking.
- State preservation.
- Security and access control.

How do we maintain a user state b/w multiple request in a stateless protocol like HTTP.
→ Session management

- Session management
  - It is a mechanism that allow a web application to create, store maintain & validate the state of the user across multiple HTTP requests.

- Cloud.
  - AWS
  - GCP
  - Microsoft Azure

- Service provided.
  - SaaS.
  - IaaS. → EC2. (Elastic compute Cloud).
    - We choose the OS
    - We choose the Instance size (RAM/CPU)
    - We configure storage
    - We install dependencies
    - We deploy our application manually.
  - PaaS
    - Directly provide you the Infrastructure
    - Here, we simply package our code and push it.

EBS
Elastic Beanstalk.
- Here we dont need to configure anything.

Choosing b/w IaaS & PaaS.

IaaS (EC2) → good understanding of RAM/CPU/OS. and want cost optimization

PaaS (Elastic beanstalk) → faster deployment with less Infrastructure

AWS provides a wide variety of services
- EC2.
- EBS.
- S3. (Simple Storage Service) → Object storage.
- SageMaker → managed machine learning platform → PaaS.
- RDS (Relational Database Service) → managed database hosting.
- Redshift → Data warehousing Services

S3

Persisted storage

IAM (Identity and access management).

↳ define who can do what on which AWS resource

Deployment flow recap up to this point

- Create an AWS account
- Rent a server → EC2.
- Transfer project files to the cloud server
  ↳ scp.
- Configure and run the application on the remote machine
  ↳ SSH.
  secure shell.